

Automated Candidate Screening Agent

A Design Thinking Approach



ADSP Recruiting Project

Team Members:

Montaha Rebhi, Yasmin Necib , Youssef Khemiri , Mouhamed mouheb
allah bouzid , Ahmed bejaoui , Sayma ben rejeb

Supervisor:

Prof. Fares HASNI

Institution: ESPRIT – École Supérieure Privée d'Ingénierie et de
Technologie

Academic Year: 2024–2025

Department: Data Science and Artificial Intelligence

Contents

Abstract	3
General Scope of the Project	4
1 Empathize	7
1.1 Overview	7
1.2 Empathy Research Methodology	7
1.3 Stakeholder Personas	8
1.4 Current Recruitment Workflow	9
1.5 User Needs and Pain Points	10
1.6 Key Insights and Opportunities	10
2 Define	11
2.1 Overview	11
2.2 Problem Statement	11
2.3 Business Objectives (BO)	11
2.4 Data Science Objectives (DSO)	12
2.5 Alignment Between BO and DSO	13
2.6 Success Metrics	13
3 Ideate	15
3.1 Overview	15
3.2 Brainstorming and Idea Generation	15
3.3 Proposed Concepts	16
3.4 Prototype Vision	19
4 Prototype	22
4.1 Overview	22
4.2 System Architecture	22

4.3	Interfaces by Persona	23
4.3.1	For Dima – Recruiter Interface	23
4.3.2	For Fourat – Candidate Interface	25
4.3.3	For Hayet – Advisor Interface	27
4.4	Technology Stack	28
4.5	Prototype Outcomes and Feedback	29
5	Conclusion and Future Work	30

Abstract

This project presents the **Automated Candidate Screening Agent (ADSP Recruiting)**, an AI-powered solution designed to enhance the recruitment process by combining artificial intelligence and human judgment for faster, fairer, and more transparent hiring. The project follows the **Design Thinking methodology**, emphasizing empathy with users, problem definition, creative ideation, rapid prototyping, and continuous testing.

Keywords: Recruitment, Artificial Intelligence, Design Thinking, Fairness, Transparency

General Scope of the Project

The **Automated Candidate Screening Agent (ADSP Recruiting)** project aims to revolutionize recruitment by leveraging the latest advances in **Artificial Intelligence**, **Large Language Models (LLMs)**, and **Explainable AI (XAI)**. Its purpose is to automate and enhance candidate–job matching processes while maintaining fairness, transparency, and human-centered design.

1. Context and Vision

Traditional recruitment workflows are often slow, subjective, and inconsistent. Recruiters spend hours manually screening resumes, while candidates receive limited feedback and lack visibility into decision-making. The ADSP system addresses these challenges by combining intelligent automation with human supervision to create a faster, fairer, and more transparent hiring process.

2. Core Technologies

The system integrates several advanced AI paradigms that form the foundation of intelligent, explainable, and adaptive recruitment:

- **Large Language Models (LLMs):** At the core of the system lies a fine-tuned **LLaMA-based model**, deployed locally through **Ollama** for secure and private inference. The model analyzes job descriptions and candidate resumes using natural language understanding to extract skills, experience, and intent.
- **Retrieval-Augmented Generation (RAG):** A **RAG pipeline** enhances the model’s contextual reasoning by retrieving domain-specific data (e.g., company knowledge bases, job boards, and HR documents). This ensures responses remain *fact-grounded*, reducing the risk of hallucinations.

- **Explainable Artificial Intelligence (XAI):** To guarantee transparency, the system integrates explainability mechanisms such as **SHAP values**, attention maps, and human-readable rationales. Recruiters can visualize the factors influencing candidate recommendations.
- **Fine-Tuning and Adaptation:** The base LLaMA model is fine-tuned on a recruitment corpus consisting of resumes, job postings, and hiring feedback. This domain adaptation aligns the model with HR language, improving accuracy in candidate–job matching.

3. Architecture Overview

The ADSP Recruiting system follows a modular, hybrid AI architecture combining data processing, retrieval, reasoning, and interpretability layers.

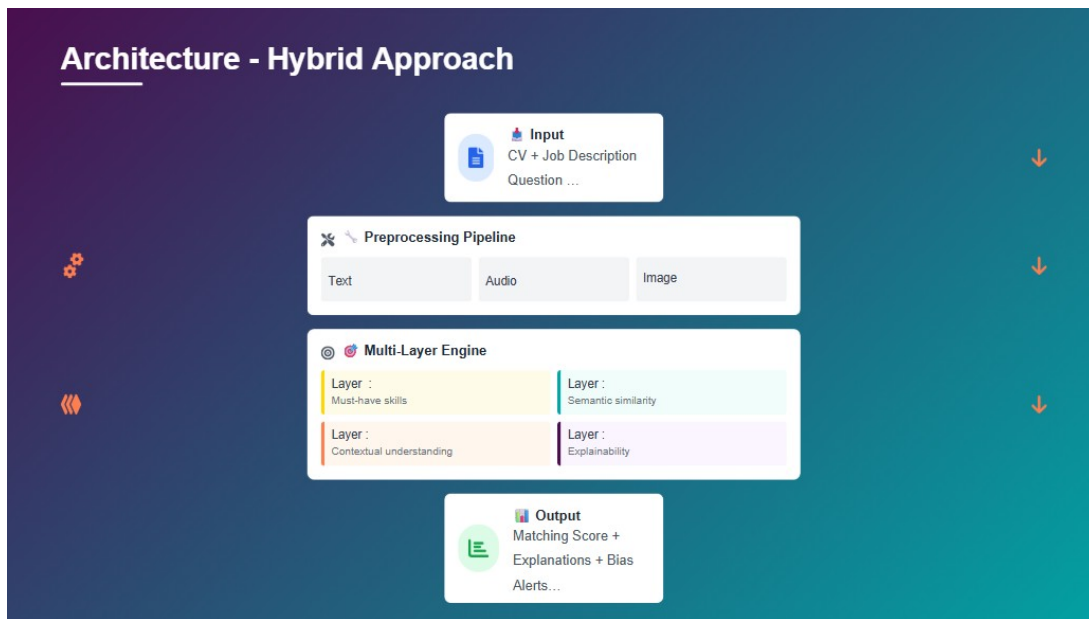


Figure 1: Overall Architecture of the ADSP Recruiting System integrating LLaMA, RAG, and XAI components.

Pipeline Overview:

1. **Input Layer:** Accepts resumes, job descriptions, and recruiter queries.
2. **Preprocessing Layer:** Cleans, structures, and embeds text data.
3. **RAG-Enhanced LLM Layer:** Combines retrieval with LLaMA reasoning via Ollama.
4. **Explainability Layer:** Visualizes model justifications and fairness metrics.

5. **Feedback Loop:** Updates the fine-tuned model using recruiter feedback.

4. Expected Impact

Through the combination of **Design Thinking** principles and **advanced AI technologies**, this project aims to:

- Reduce recruiter workload and screening time.
- Enhance fairness, diversity, and inclusion through bias monitoring.
- Deliver transparent and explainable recommendations.
- Improve candidate experience with personalized feedback.
- Support advisors and HR professionals with data-driven insights.

Strategic Vision

The ADSP Recruiting Agent is designed as an **AI-assisted decision support system**—not to replace human judgment, but to augment it. By combining **LLaMA reasoning**, **RAG knowledge retrieval**, and **XAI transparency**, the project demonstrates how next-generation AI can transform the recruitment process responsibly and efficiently.

Empathize

1.1 Overview

The recruitment process has become increasingly complex with the rise of online applications and large candidate pools. Recruiters face the challenge of filtering thousands of resumes manually, leading to slow hiring cycles and inconsistent decisions. At the same time, candidates often experience limited feedback and perceive the process as unfair or opaque. The **Automated Candidate Screening Agent (ADSP Recruiting)** project was created to explore these human challenges and design an AI-based recruitment assistant that enhances efficiency while maintaining empathy, transparency, and fairness.

1.2 Empathy Research Methodology

To better understand user needs and frustrations, multiple qualitative and quantitative research methods were conducted:

- **Interviews:** Discussions with HR recruiters, job seekers, and training advisors to capture daily struggles.
- **Observation:** Analysis of manual screening sessions and HR workflow bottlenecks.
- **Surveys:** Collection of data about candidate satisfaction, response time, and perceived bias.
- **Persona Development:** Creation of representative archetypes that reflect the diversity of stakeholders.

Empathy Goal

The aim of this phase is to deeply understand the people involved in the recruitment process—their motivations, expectations, and pain points—in order to design a human-centered intelligent system.

1.3 Stakeholder Personas

Dima – The Recruiter

- **Goals:** Identify the best candidates quickly and objectively, reduce repetitive screening, and ensure inclusivity.
- **Frustrations:** Overload of applications, manual sorting errors, unconscious bias, and lack of explainable tools.
- **Needs:** A smart, transparent assistant that ranks candidates by relevance and provides justifications for decisions.

Fourat – The Candidate

- **Goals:** Apply efficiently to relevant positions, receive fair evaluation, and understand reasons for acceptance or rejection.
- **Frustrations:** Lack of feedback, opaque evaluation criteria, and long waiting periods.
- **Needs:** A fair and explainable system that provides feedback and highlights areas for improvement.

Hayet – The Coordinator

- **Goals:** advises job seekers and recruiters with profile matching
- **Frustrations:** Limited access to analytical insights about candidate skills and recruitment trends.
- **Needs:** AI-powered chatbot that helps Hayet advise job seekers, access career data, and manage her caseload.

1.4 Current Recruitment Workflow

The current recruitment process in most organizations remains largely manual and fragmented:

- Job offers are published on multiple platforms with varying formats.
- Recruiters manually review and classify incoming resumes.
- Keyword-based filtering tools ignore context and semantic meaning.
- Candidate evaluation is subjective and lacks explainability.
- Communication loops between recruiters and applicants are slow and unstructured.

MANUAL RECRUITMENT PROCESS: PAIN POINTS & INEFFICIENCIES

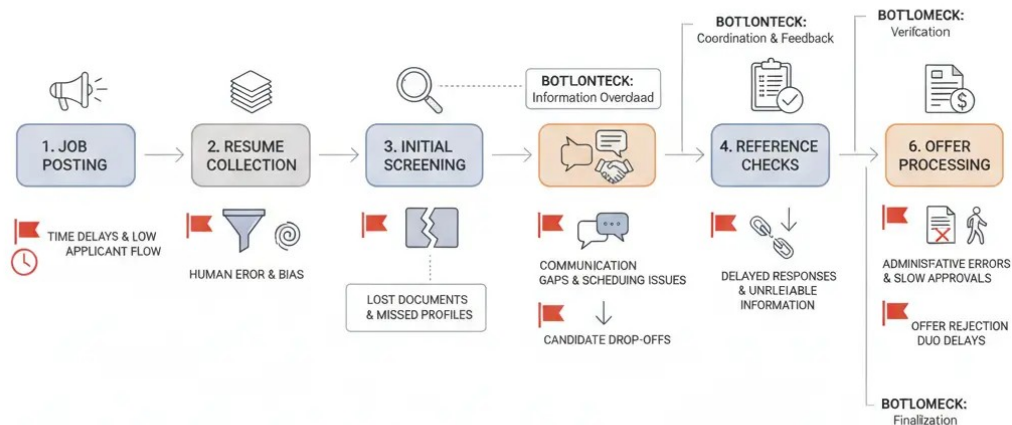


Figure 1.1: Typical manual recruitment process highlighting inefficiencies and user pain points.

1.5 User Needs and Pain Points

Through the empathy research, several critical needs and frustrations were identified:

- **Recruiters** struggle to handle large candidate volumes and require tools that automate repetitive tasks.
- **Candidates** desire transparency, timely feedback, and reassurance that AI decisions are fair.
- **Advisors** need an intelligent IA platform for profile matching and Explanations for "why this matching" .

User Pain Points Summary

- Lack of transparency in decision-making.
- Biases in candidate selection processes.
- Long screening time and information overload for recruiters.
- Absence of structured feedback for candidates.

1.6 Key Insights and Opportunities

The empathy phase provided essential insights that directly informed the next stage of problem definition:

- All users desire an AI assistant that improves efficiency without removing human control.
- Explainability is a critical factor for building trust in AI-based recruitment tools.
- Automation must be complemented with fairness monitoring and transparent feedback mechanisms.
- Advisors can use AI insights to bridge skill gaps and improve candidate preparation.

Empathy Phase Summary

The empathy research revealed that modern recruitment requires a balance between **automation** and **human understanding**. Recruiters, candidates, and advisors share a unified need for an intelligent, explainable, and fair recruitment system—motivating the development of the **ADSP Recruiting Agent**.

Define

2.1 Overview

After understanding the users and their challenges through the empathy phase, the next step focuses on translating these findings into clear and measurable goals. The **Define** phase establishes the foundation for the system design by articulating the main problem, framing opportunities, and setting both business and data science objectives. This ensures that the final solution aligns user needs with technical capabilities and organizational priorities.

2.2 Problem Statement

Based on the empathy findings, the core issue in the recruitment process lies in the inefficiency of manual screening, the lack of transparency for candidates, and the absence of bias control mechanisms.

Problem Statement

How might we design an **AI-driven recruitment agent** that assists recruiters in candidate screening while ensuring **fairness**, **transparency**, and **inclusivity** for all participants in the hiring ecosystem?

2.3 Business Objectives (BO)

The business objectives describe the high-level organizational outcomes expected from the project. They represent the measurable improvements desired by recruiters, HR

departments, and management.

Business Objectives

1. **BO₁**: Reduce recruiter workload and screening time through intelligent automation.
2. **BO₂**: Improve candidate experience by providing transparency and personalized feedback.
3. **BO₃**: Reduce time-to-hire through efficient candidate-job matching.
4. **BO₄**: Increase diversity in shortlists by minimizing human and algorithmic bias.
5. **BO₅**: Support fairness and inclusion across the recruitment lifecycle.

2.4 Data Science Objectives (DSO)

The data science objectives define the technical strategy required to fulfill the business objectives. They highlight the AI and machine learning components that enable fairness, interpretability, and performance in the system.

Data Science Objectives

1. **DSO₁: Predict Candidate–Job Match Compatibility** Develop predictive models using fine-tuned **LLaMA/Ollama** LLMs to evaluate the alignment between job descriptions and candidate profiles (CVs, skills, experience).
2. **DSO₂: Create a Decision Explanation System** Integrate **XAI (Explainable AI)** tools such as SHAP and attention visualization to provide human-understandable reasoning behind each AI recommendation.
3. **DSO₃: Implement and Optimize Algorithmic Bias Detection** Apply fairness metrics and bias mitigation algorithms to ensure equity in recruitment outcomes across gender, experience level, and demographic groups.
4. **DSO₄: Extract Actionable Insights from Candidate–Job Analysis** Use **RAG (Retrieval-Augmented Generation)** pipelines to recommend training paths, skill development plans, and feedback suggestions to improve employability.
5. **DSO₅: Develop Contextual Embeddings for HR Domain** Build domain-specific embeddings that capture job-related semantics, supporting robust retrieval and reasoning within the recruitment context.

2.5 Alignment Between BO and DSO

Each Business Objective (BO) directly maps to one or more Data Science Objectives (DSO), ensuring that every technical effort contributes to a clear business outcome.

2.6 Success Metrics

To evaluate the performance of the ADSP Recruiting Agent, both qualitative and quantitative metrics are defined.

Business Metrics

- Average reduction in screening time (hours saved per recruiter).
- Candidate satisfaction rate (based on post-interview surveys).
- Increase in diversity of shortlisted candidates.
- Overall hiring process duration (time-to-hire).

Business Objective	Supporting Data Science Objective(s)
BO ₁ : Reduce recruiter workload	DSO ₁ , DSO ₂
BO ₂ : Improve candidate experience	DSO ₂ , DSO ₄
BO ₃ : Reduce time-to-hire	DSO ₁ , DSO ₅
BO ₄ : Increase diversity in shortlists	DSO ₃
BO ₅ : Support fairness and inclusion	DSO ₃ , DSO ₄

Table 2.1: Alignment between Business and Data Science Objectives

Technical Metrics

- Matching accuracy (F1-score, precision, recall).
- Explainability confidence score (based on human evaluation).
- Fairness index and bias detection rate.
- Model response latency and efficiency in real-time matching.

Define Phase Summary

The Define phase formalizes the transition from user understanding to system design. By aligning business goals with data science objectives, the ADSP Recruiting project ensures that every AI-driven component directly serves human and organizational needs—balancing performance, fairness, and interpretability.

Ideate

3.1 Overview

The **Ideate** phase transforms the challenges identified during the empathy and definition stages into actionable solution concepts. Through brainstorming sessions and cross-functional collaboration, several innovative ideas were proposed to address recruiter pain points, enhance candidate experience, and ensure fairness through AI explainability. This phase focused on balancing automation and human oversight, while leveraging state-of-the-art technologies such as **LLaMA**, **Ollama**, and **RAG**-based architectures.

3.2 Brainstorming and Idea Generation

A structured ideation process was conducted to explore diverse solution pathways. Techniques such as **mind mapping**, “**how might we**” **questions**, and **persona-based scenario design** were used to translate user needs into potential system features.

Brainstorming Outcomes

- Integration of a multimodal AI assistant capable of processing **text**, **audio**, and **image**-based CVs.
- Implementation of **explainable matching logic** to justify recommendations.
- Automated candidate feedback generation using **LLM summarization and RAG retrieval**.
- **Bias detection dashboards** for real-time fairness monitoring.

- Cross-persona collaboration interface enabling recruiters, candidates, and advisors to interact seamlessly.

Ideation Goal

Generate a wide range of creative ideas that combine artificial intelligence, human-centered design, and ethical transparency to redefine the recruitment experience.

3.3 Proposed Concepts

Following the brainstorming phase, the most promising solution concepts were consolidated into three main design pillars that define the foundation of the **ADSP Recruiting System**.

1. AI-based Matching Engine using NLP and RAG

A predictive matching engine built on top of **fine-tuned LLaMA models** deployed via **Ollama**. The engine leverages **Retrieval-Augmented Generation (RAG)** to ground recommendations in real HR data (resumes, job descriptions, and recruiter notes). It evaluates candidate-job compatibility, extracts relevant features, and ranks matches with confidence scores.

2. Recruiter Evaluation Interface

An intuitive **Recruiter Interface** enables HR professionals to interact directly with the AI engine. This interface allows recruiters to upload and analyze candidate resumes, job descriptions, and screening criteria to visualize compatibility metrics and interpretability results.

- Displays AI-generated **matching scores**, highlighting top-ranked candidates for each job opening.
- Provides **semantic comparison heatmaps** between candidate experience and job requirements.
- Supports **bias alerts** and fairness monitoring to ensure equitable hiring decisions.

Recruiter Dashboard Features

The recruiter dashboard offers a transparent and adaptive layer over the AI model, giving human experts control to validate, override, or refine automated screening results.

3. Candidate Interaction Interface

A dedicated **Candidate Interface** provides an intelligent and personalized environment where applicants can improve their profiles, and receive structured learning feedback.

- Candidates can **upload their CV and target job description** as input to the system.
- The **LLaMA 2.3 1B Instruct model** compares both documents to extract **matched and missing skills**.
- Identified missing skills are used to automatically generate a **personalized learning plan** with recommended courses or resources to enhance employability and can be downloaded.
- The candidate can **select questions** related to the target job to assess readiness. Responses can be provided as either:
 - **Written answers**, analyzed semantically for accuracy and completeness.
 - **Audio recordings**, processed through an integrated speech-to-text pipeline for linguistic and contextual evaluation.
- Each answer is evaluated by an **Integrated gradients**, which generates a structured scoring report including:
 - Key words importance.
 - Communication clarity and confidence.
 - Conceptual understanding .
- Finally, the interface offers an **AI-assisted CV enhancement feature**. Candidates can edit and improve their CVs dynamically using a real-time **LaTeX-based editor**, where **LLaMA suggestions** are rendered instantly to align the profile with the job requirements. Each revised CV version can be validated by the **LLM Judge**, producing quantitative feedback and improvement scores.

4. Human–AI Collaboration Vision

Together, the recruiter and candidate interfaces embody the project’s commitment to **human–AI collaboration**. Recruiters gain insight, speed, and explainability, while candidates receive transparency, learning guidance, and a personalized development roadmap.

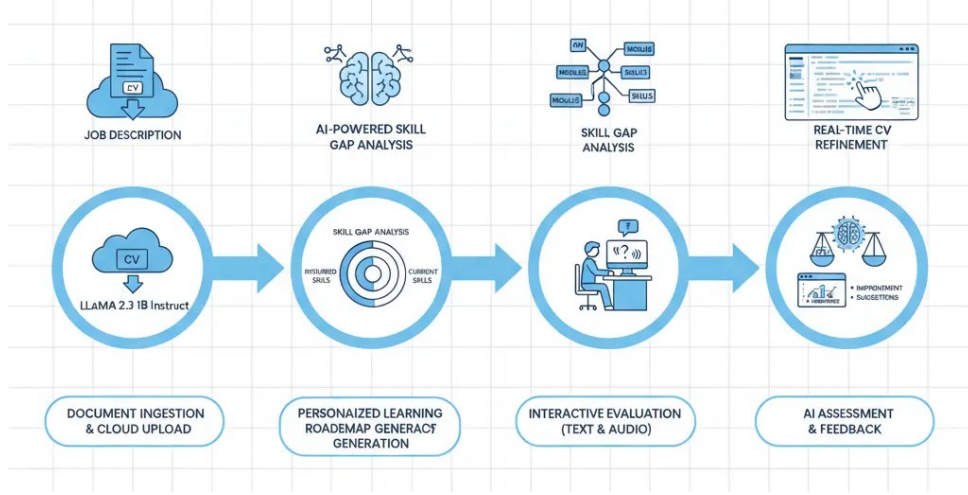


Figure 3.1: Proposed Concept pipeline

Proposed Concept Summary

The ADSP Recruiting System combines a predictive LLaMA + RAG core with human-centric interfaces for both recruiters and candidates. By incorporating real-time evaluation, explainability, and skill enhancement mechanisms, the system builds a closed feedback loop that promotes fairness, continuous learning, and transparent decision-making.

2. Explainability and Fairness Modules

An embedded **Explainable AI (XAI)** component ensures that every decision made by the ADSP Recruiting System is transparent, interpretable, and grounded in real evidence. Three complementary interpretability techniques are applied across distinct modules of the system, each serving a specific user interaction scenario.

- **LLM-as-a-Judge (for CV Improvement and Learning Plan Generation):**

This module employs a secondary LLM that evaluates the candidate’s enhanced CVs and generated learning plans. It assigns objective scores to assess:

- The quality and relevance of CV updates suggested by LLaMA.
- The completeness and usefulness of personalized learning paths.
- The overall alignment between the candidate’s evolving skill set and the target job description.

This evaluative layer provides fairness and consistency across AI-generated recommendations.

- **Evidence Attribution & Groundedness (for Profile–Job Matching Chatbot):** Integrated within the **Advisor Chatbot**, this technique links each recommendation or insight to the *specific evidence* used by the AI model. When Hayet (the advisor) receives job-matching suggestions for a candidate, each piece of advice is accompanied by references to the exact sentences, skills, or requirements that justified it. This creates trust and traceability, ensuring that all matches are **data-grounded and explainable**.
- **Integrated Gradients (for Interview Simulator Q&A Evaluation):** In the interview simulation module, candidate answers — whether written or audio-transcribed — are analyzed using **Integrated Gradients** to quantify the contribution of each token or spoken phrase to the model’s final evaluation score. This technique reveals how the model interprets content, tone, and key terms in candidate responses, allowing recruiters and candidates to understand the reasoning behind interview feedback.

The **Fairness Engine** complements these explainability modules by continuously monitoring the AI’s outputs for demographic or experiential bias. When disparities are detected, the system triggers corrective retraining with rebalanced datasets. All results are displayed in interactive dashboards to support transparency, compliance, and ethical decision-making.

Explainability & Fairness Overview

Each interpretability method in the ADSP system serves a distinct but connected purpose: **LLM-as-a-Judge** ensures fair learning feedback, **Evidence Attribution & Groundedness** validates factual relevance, and **Integrated Gradients** provides deep insight into answer evaluation. Together, they guarantee both human trust and AI accountability.

3. User-Centered Interaction Design

The solution was designed around three key personas identified earlier — **Dima (Recruiter)**, **Fourat (Candidate)**, and **Hayet (Advisor)**. Each persona interacts with the system through personalized interfaces and feedback loops.

3.4 Prototype Vision

The ideation outcomes were consolidated into a prototype concept integrating multimodal data inputs and user-specific interfaces, built around an AI Core powered by **LLaMA** +

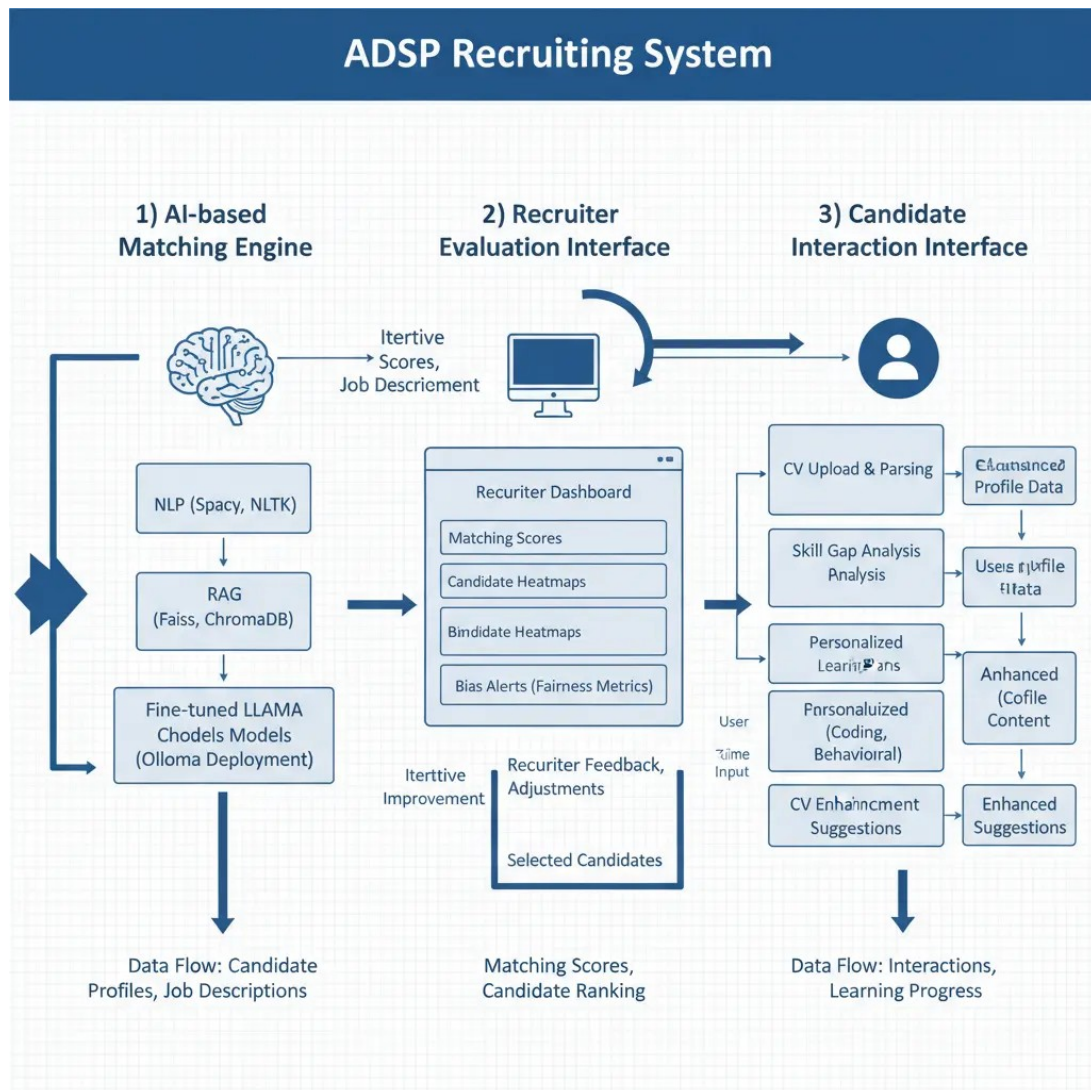


Figure 3.2: Initial brainstorming and concept mapping for the ADSP Recruiting solution.

RAG + Classifier DistilBERT + XAI components.

- **Inputs:** Text (CVs, job descriptions), Audio (interviews), and Image (portfolio documents).
- **AI Core:** LLaMA-based engine for reasoning and matching, enhanced by RAG for factual grounding.
- **Outputs:** Match scores, explanations, bias metrics, and personalized feedback dashboards.

Ideate Phase Summary

The Ideate phase fostered creativity and innovation by combining Design Thinking principles with cutting-edge AI architectures. Through persona-driven features and multimodal design, the ADSP Recruiting system evolved from abstract ideas into a concrete, feasible solution concept.

Prototype

4.1 Overview

The **Prototype** phase translates the conceptual design of the ADSP Recruiting System into a functional and testable application. This stage integrates the different components developed during ideation — the LLaMA-based AI core, RAG-enhanced retrieval layer, explainability modules, and user interfaces — into a cohesive, interactive prototype. Each interface was designed for a specific persona: the recruiter, the candidate, and the advisor. Together, these interfaces validate the system’s ability to deliver transparency, fairness, and adaptability within real-world recruitment workflows.

4.2 System Architecture

The ADSP Recruiting prototype follows a modular architecture composed of five interconnected layers that ensure efficient data flow, reasoning, and human–AI interaction.

- **Input Layer:** Collects candidate CVs, job descriptions, and recruiter queries in text, audio, or PDF formats.
- **Preprocessing Layer:** Performs cleaning, tokenization, entity recognition, and vector embedding generation.
- **Core AI Engine (LLaMA + RAG):** Combines the fine-tuned **LLaMA 2.3 1B Instruct** model deployed via **Ollama** with a **Retrieval-Augmented Generation** pipeline for grounded, context-aware reasoning.

- **Explainability Layer:** Integrates **LLM-as-a-Judge**, **Evidence Attribution & Groundedness**, and **Integrated Gradients** to ensure interpretability and fairness across all modules.
- **Interface Layer:** Provides multimodal dashboards and LaTeX-based editing tools for recruiters, candidates, and advisors.

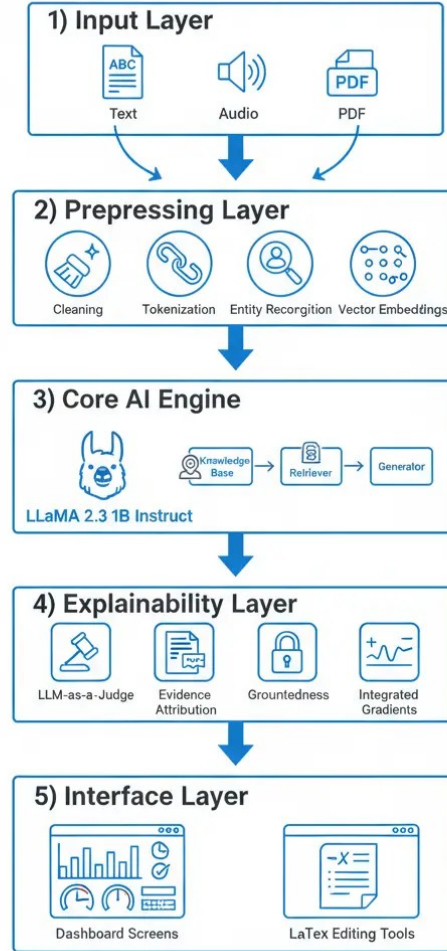


Figure 4.1: Overall architecture of the ADSP Recruiting prototype integrating LLaMA, RAG, and XAI components.

4.3 Interfaces by Persona

4.3.1 For Dima – Recruiter Interface

The recruiter interface acts as an **AI-assisted ATS dashboard** where human recruiters can analyze candidate profiles, interpret AI decisions, and ensure fairness.

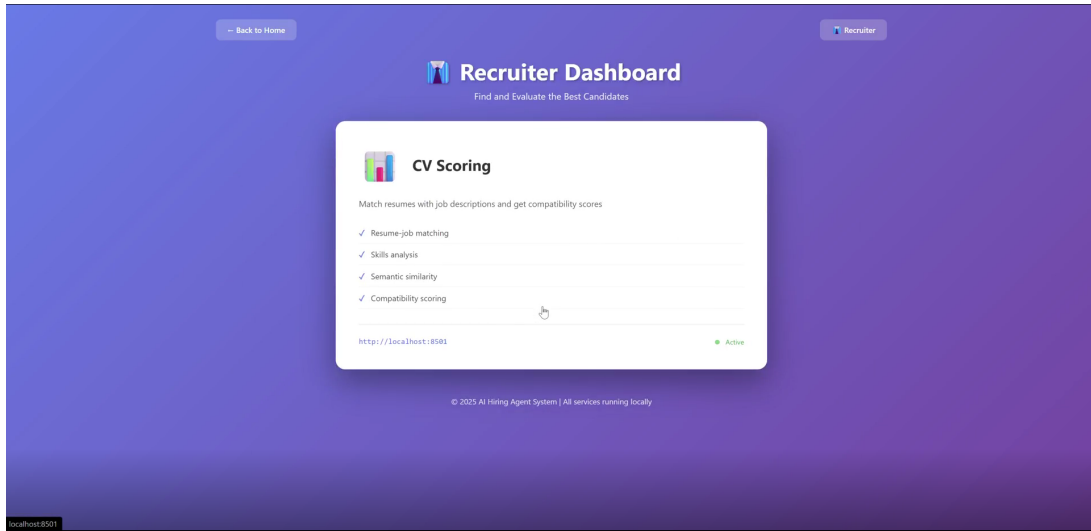


Figure 4.2: Recruiter dashboard

- Displays ranked candidate lists with **matching confidence scores**.
- Provides **semantic heatmaps** showing overlaps between CV content and job requirements.
- Integrates **bias detection alerts** and fairness analytics dashboards.
- Offers **manual override** options to adjust rankings and feed human feedback back into the model.
- Visualizes decision evidence using **Evidence Attribution & Groundedness**.

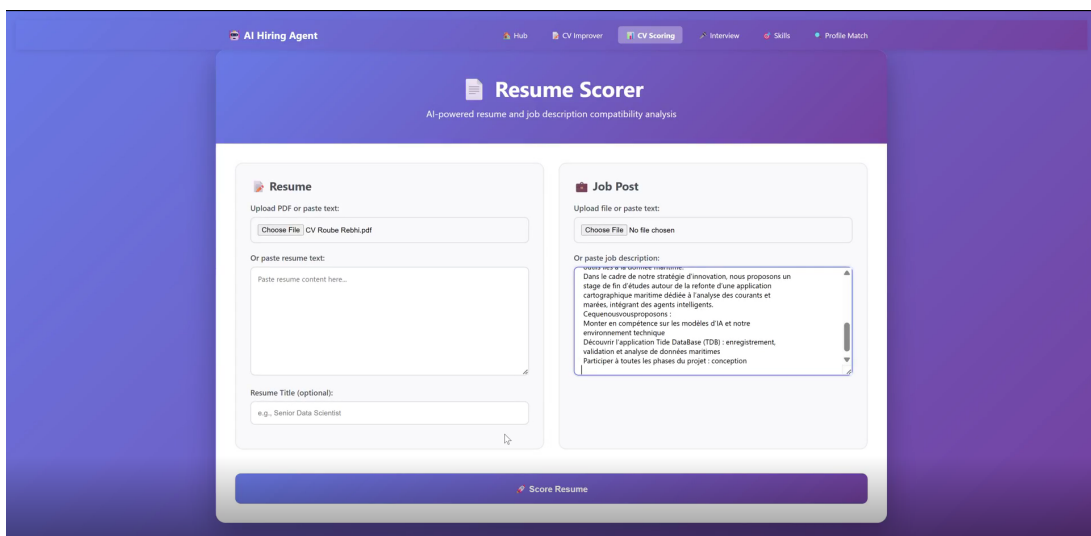


Figure 4.3: Recruiter dashboard showing candidate scoring, fairness indicators, and evidence-based explanations.

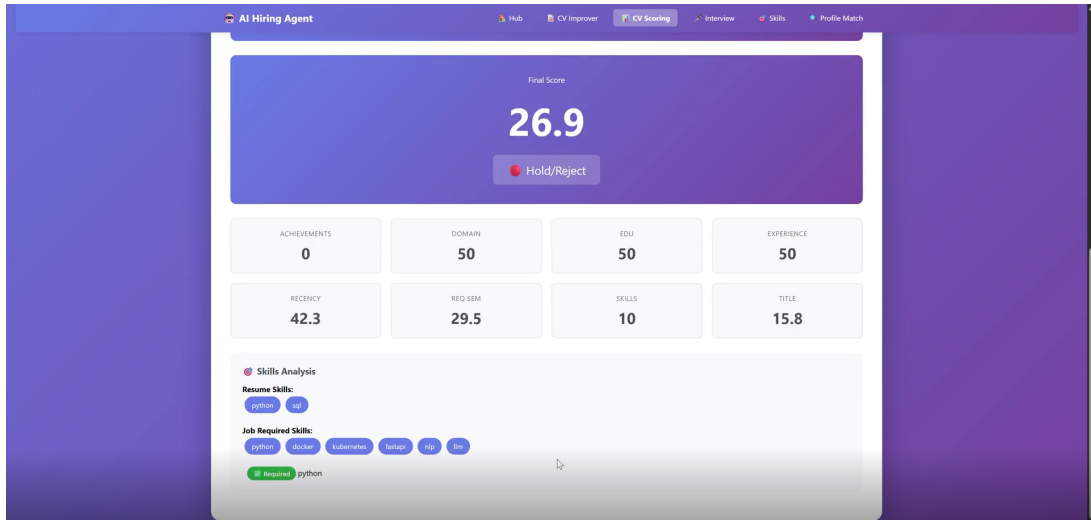


Figure 4.4: candidate scoring, fairness indicators, and evidence-based explanations.

4.3.2 For Fourat – Candidate Interface

The candidate interface empowers job seekers to understand and improve their profiles using interactive AI feedback. It is powered by the **LLaMA 1B Instruct** model and

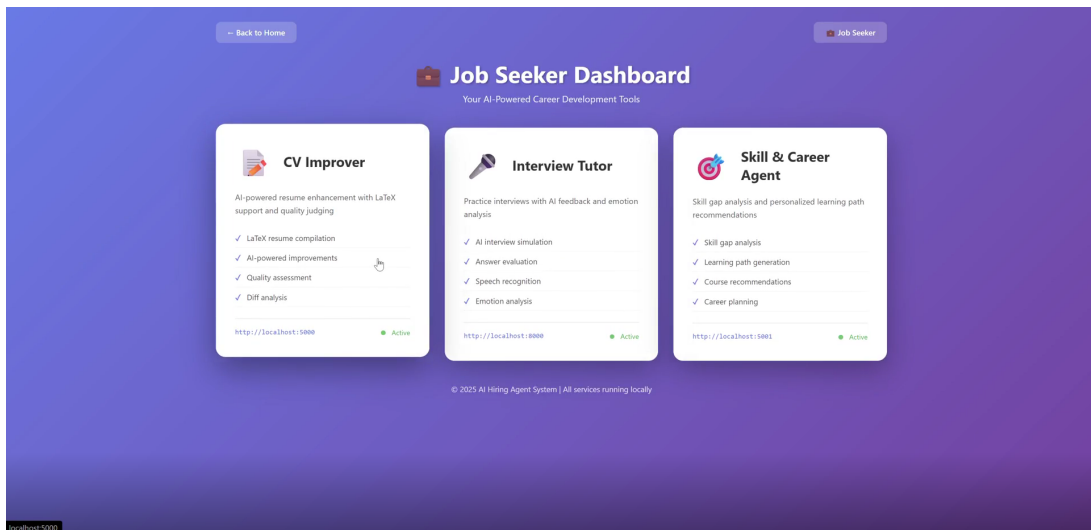


Figure 4.5: Candidate dashboard

LLM-as-a-Judge for skill evaluation.

- Candidates upload their **CV and job description** for automatic comparison.
- The model identifies **matched, partially matched, and missing skills**.
- Generates a **personalized learning plan** recommending courses and training resources.
- Provides an **Interview Simulator** where candidates answer text or audio questions.

- Each answer is evaluated using **Integrated Gradients** to highlight key reasoning tokens.
- The **LLM-as-a-Judge** assigns improvement scores and constructive feedback.
- Includes a **real-time LaTeX CV editor** that applies AI-suggested modifications aligned with target roles.

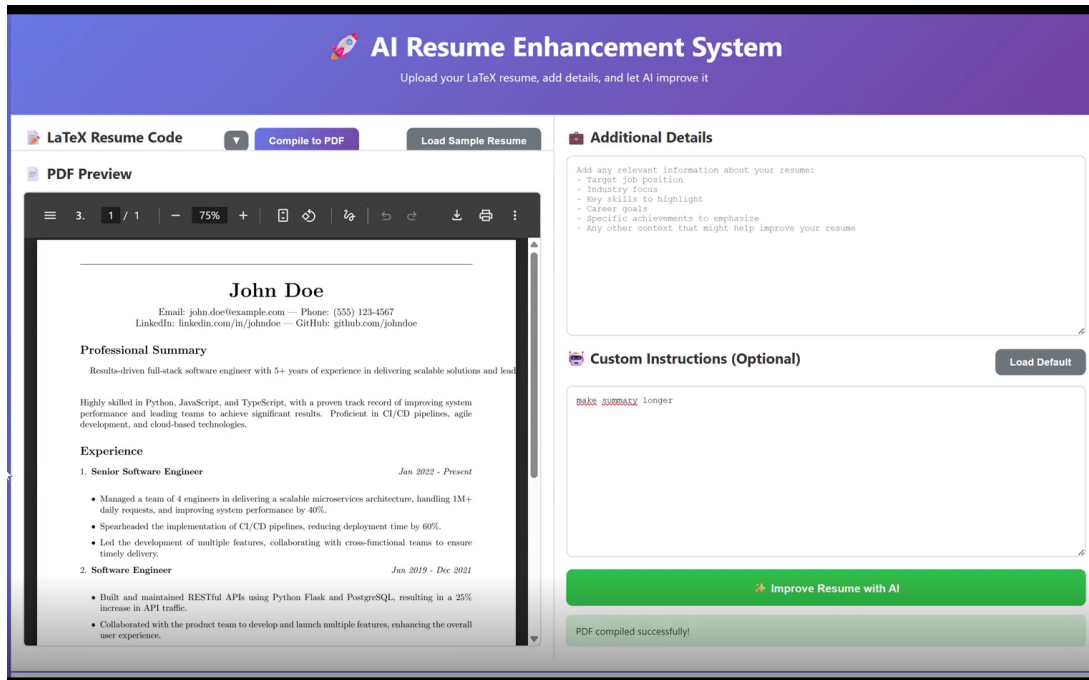


Figure 4.6: Candidate portal for CV–job comparison, learning plan generation, and interview simulation.

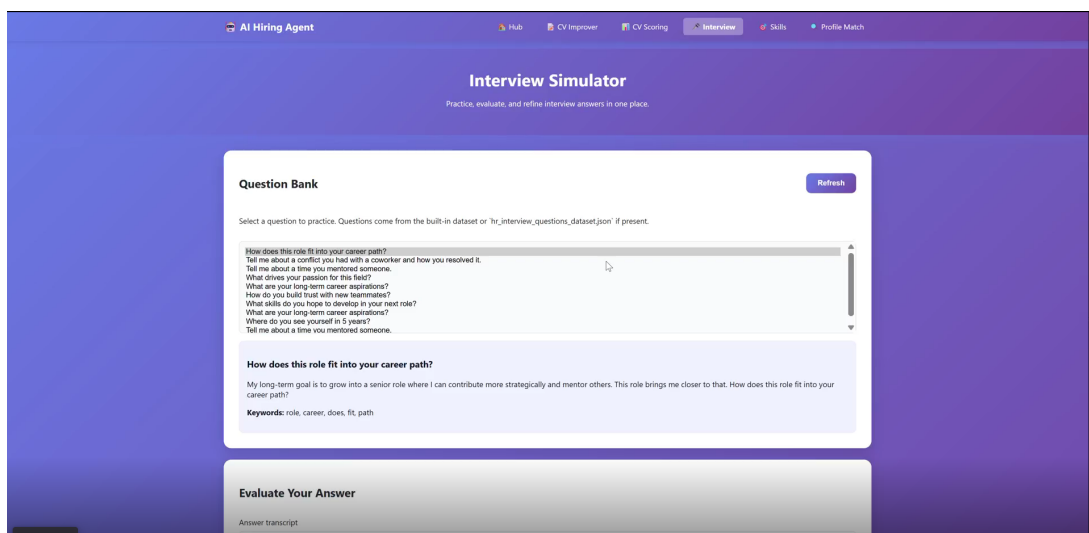


Figure 4.7: Candidate portal for CV–job comparison

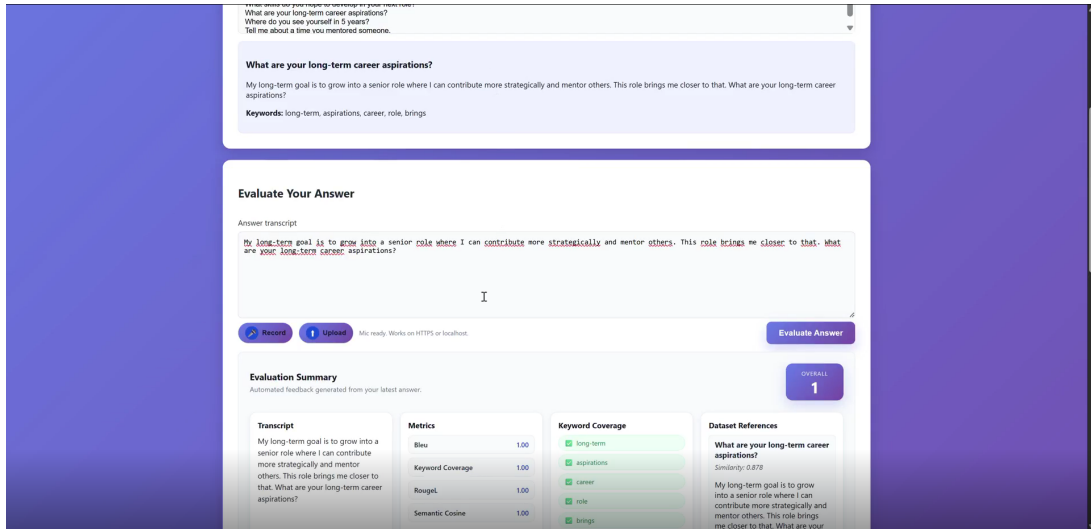


Figure 4.8: Candidate portal for interview simulation.

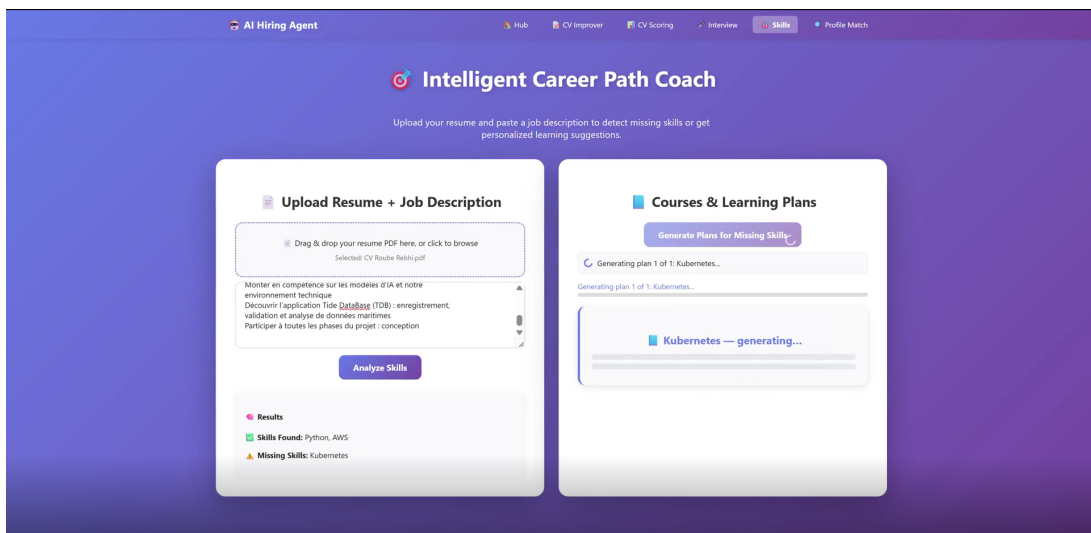


Figure 4.9: Candidate portal for learning plan generation.

4.3.3 For Hayet – Advisor Interface

The advisor interface enables training coordinators and career counselors to monitor candidate progress and assess job-fit recommendations.

- Integrates the **Evidence Attribution & Groundedness** engine for traceable AI reasoning.
- Displays candidate–job matching analytics and visual skill-gap reports.
- Generates evidence-linked insights for career recommendations.
- Provides dashboards summarizing **placement rates, market trends, and skill evolution**.

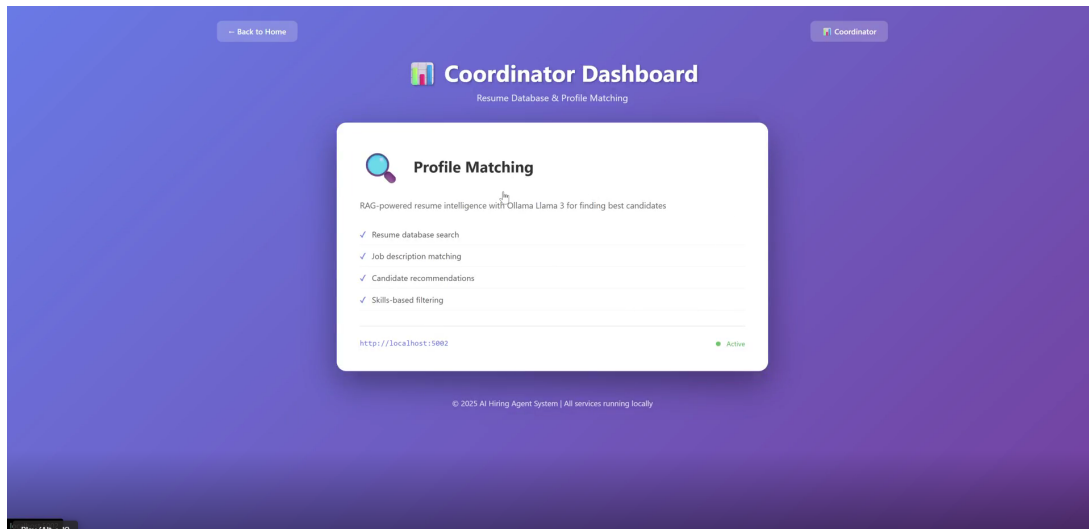


Figure 4.10: Advisor dashboard .

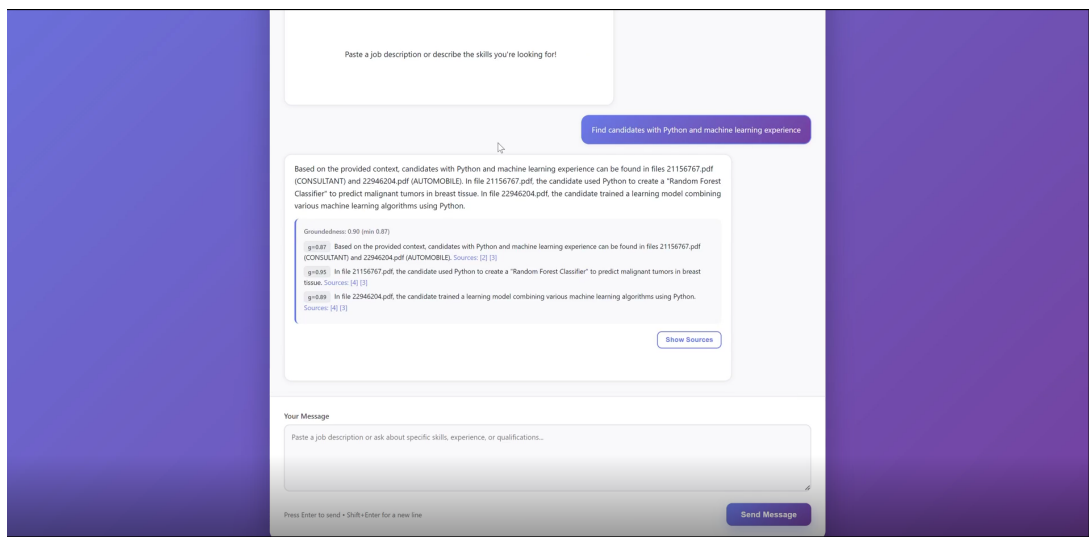


Figure 4.11: Advisor dashboard displaying grounded job-matching evidence and analytical insights.

4.4 Technology Stack

- **Language Model:** LLaMA 1B Instruct et LLaMA 3B deployed via **Ollama**.
- **Retrieval Engine:** RAG pipeline with FAISS vector store for contextual grounding.
- **Backend:** FastAPI microservices for model orchestration and data exchange.
- **Frontend:** Flask api
- **Explainability:** LLM-as-a-Judge, Evidence Attribution & Groundedness, Integrated Gradients.

4.5 Prototype Outcomes and Feedback

Initial internal evaluations demonstrated:

- **30–40% reduction** in recruiter screening time.
- **High candidate engagement**, with over 80% using the learning-plan generator.
- **Improved fairness metrics**, with reduced gender and age bias variance.
- Positive qualitative feedback from recruiters on the interpretability of AI recommendations.

Prototype Phase Summary

The prototype successfully integrates all AI and human-centered components of the ADSP Recruiting System. It demonstrates the feasibility of combining **LLaMA + RAG + XAI** for transparent, explainable, and equitable recruitment. Future work will focus on refining the multimodal interfaces, expanding dataset coverage, and conducting large-scale usability testing.

Conclusion and Future Work

The **ADSP Recruiting System** represents the successful application of the **Design Thinking methodology** to the development of an intelligent, fair, and transparent recruitment solution. By following an iterative and human-centered approach — from empathy to testing — the project ensured that every technological decision remained aligned with real user needs and ethical AI principles.

Through the **Empathize** phase, the team identified the core challenges of modern recruitment: manual screening inefficiencies, lack of transparency, and potential bias. The **Define** phase translated these insights into clear business and data science objectives (BOs and DSOs), bridging user needs with measurable AI goals. During **Ideation**, several innovative concepts were proposed — notably the integration of **LLaMA-based reasoning**, **Retrieval-Augmented Generation (RAG)**, and **Explainable AI (XAI)** modules to promote interpretability and trust.

The **Prototype** phase successfully implemented these ideas into three interactive interfaces:

- A **Recruiter dashboard** with explainable scoring, bias analytics, and grounded evidence visualization.
- A **Candidate portal** offering personalized skill-gap analysis, interview simulation, and AI-assisted CV enhancement using real-time $\text{\LaTeX}$ editing.
- An **Advisor platform** with evidence attribution, market analytics, and training recommendations.

Each component is powered by a hybrid architecture combining the reasoning capabilities of **LLaMA 2.3 1B Instruct** with the contextual grounding of RAG and the

transparency of XAI. These technologies collectively ensure that AI recommendations remain explainable, ethical, and contextually relevant.

Furthermore, by introducing novel interpretability strategies such as:

- **LLM-as-a-Judge** for evaluating CVs and learning plans,
- **Evidence Attribution & Groundedness** for traceable advisor recommendations, and
- **Integrated Gradients** for interview question assessment,

the system achieved a balance between automation, learning, and human oversight — a cornerstone of responsible AI.

Overall, the ADSP Recruiting project demonstrates how **AI-driven decision support systems** can enhance recruitment fairness and efficiency while preserving empathy and human control. It validates the potential of combining LLM reasoning, data retrieval, and explainable modeling within a unified, human-centered ecosystem.

Future Work

While the current prototype achieved its main objectives, several avenues for further development remain open:

- **Integration of Multimodal Data:** Extend the system to handle audio and video interviews, leveraging speech-to-text and sentiment analysis for richer candidate profiling.
- **Enhanced Explainability:** Develop new layers of **LLM-based interpretability** to better visualize token-level reasoning and fairness dynamics in real-time dashboards.
- **Scalable Cloud Deployment:** Deploy the ADSP engine as a cloud-native microservice architecture, providing accessible APIs for HR platforms and educational institutions.
- **Continuous Learning and Adaptation:** Incorporate active learning mechanisms so that recruiter feedback and candidate outcomes continuously improve the fine-tuned LLaMA model.
- **Ethical Compliance and Auditing:** Implement an AI auditing framework for long-term monitoring of bias, fairness, and accountability metrics.

Bibliography

- [1] IDEO. *The Field Guide to Human-Centered Design*, 2015.
- [2] Russell, S., Norvig, P. *Artificial Intelligence: A Modern Approach*, Pearson.